Block 6 • Hybrid systems

Method-selection guide: information and cold start

Method	Main information	Cold-item support	Explainability
Content-based	metadata / text / features	strong for new items	often clear
User-kNN	interaction history	weak	medium
Item-kNN	interaction history	weak	medium
MF / implicit ALS	interaction history	weak without features	limited
Hybrid / FM	metadata + interactions	stronger	varies

Method-selection guide: project situations

Situation	Good first method	Why
New item with metadata	content-based or hybrid	metadata can represent item before interactions exist
Warm ratings dataset	item-kNN + MF	strong classical baselines
Clicks / views / purchases	implicit ALS, BPR, EASE	ranking from positive-unlabelled data
Rich metadata + interactions	FM / LightFM / hybrid	learn feature interactions and behavioural patterns
Need transparent explanation	content-based or item-kNN	easier to justify to users

Takeaway from Block 6

Keep these three ideas active for the capstone project.

1 Pure methods are rare in practice

Real recommenders combine content, interactions, context, rules, business goals and constraints.

2 Factorization Machines generalize MF

FM learns pairwise feature interactions and can include users, items, metadata and context in one feature vector.

3 Choose methods by problem conditions

Cold start, sparsity, metadata quality, latency, explainability and evaluation goals should drive model choice.

Capstone checkpoint

End-of-Day 2 project checkpoint

1. **Dataset confirmed**
   Which dataset will your group use, and what feedback signals are available?

2. **Task defined**
   Rating prediction, top-K recommendation, next-item prediction or another task?

3. **Baseline selected**
   Popularity, content-based, user-kNN or item-kNN?

4. **Improved method**
   MF, implicit ALS, BPR, hybrid or another method?

5. **Evaluation plan**
   Split, candidate set, metrics and limitations?